EC442 Macroeconomics for MRes students | Instructor: Matthias Doepke
London School of Economics | Autumn 2026

Problem Set 4

Growth Models with Human Capital

Question 1: Consider an economy in which the representative consumer has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t).$$

The physical resource constraint is given by:

$$c_t + i_t = A k_t^{\alpha} h_{1,t}^{1-\alpha}.$$

The laws of motion for physical and human capital are:

$$k_{t+1} = (1-\delta)k_t + i_t,$$
$$h_{t+1} = B(h_{2,t})^{\gamma},$$

where $0 < \gamma < 1$. The allocation of human capital is subject to the constraint $h_t = h_{1,t} + h_{2,t}$, with $h_{1,t}, h_{2,t} \geq 0$, and the initial capital stocks $k_0 > 0$ and $h_0 > 0$ are given.

1. Define a sequential-markets equilibrium for this economy.
2. Use first-order conditions to compute the steady state.
3. Show that the conditions for the first welfare theorem are satisfied (implying that we can solve a social planning problem to compute the equilibrium allocation).
4. Formulate the Bellman equation corresponding to the social planning problem in this economy.
5. Solve the value function numerically for the parameter values $\beta = 0.6$, $\alpha = 0.3$, $\delta = 0.6$, $\gamma = 0.6$, $A = 1$, $B = 2$ (these parameters approximate a calibration in which each model period corresponds to a decade). Limit the state space in each dimension to $[0.4\bar{k}, 1.5\bar{k}]$ and $[0.4\bar{h}, 1.5\bar{h}]$, where $\bar{k}$ and $\bar{h}$ are the steady state values. Provide graphs of the value and policy functions, and provide time-series graphs comparing outcomes over 10 periods in three economies: Economy A starts out in the steady state ($k_{a,0} = \bar{k}$, $h_{a,0} = \bar{h}$), Economy B starts out with the steady-state level of physical capital, but only half as much human capital ($k_{b,0} = \bar{k}$, $h_{b,0} = \bar{h}/2$), and Economy C starts out with the steady-state level of human capital, but only half as much physical capital ($k_{c,0} = \bar{k}/2$, $h_{c,0} = \bar{h}$). Provide economic intuition why shortfalls of physical versus human capital lead to qualitatively different transition paths.

Question 2: We now turn to a version of the Lucas (1988) model, in which human capital can be accumulated under constant returns. The representative consumer still has preferences:

$$\sum_{t=0}^{\infty} \beta^t \log(c_t).$$

The physical resource constraint is given by:

$$c_t + i_t = A k_t^{\alpha} h_{1,t}^{1-\alpha}.$$

The laws of motion for physical and human capital are:

$$\begin{aligned} k_{t+1} &= (1-\delta)k_t + i_t, \\ h_{t+1} &= B h_{2,t}. \end{aligned}$$

The allocation of human capital is subject to the constraint $h_t = h_{1,t} + h_{2,t}$, with $h_{1,t}, h_{2,t} \geq 0$, and the initial capital stocks $k_0 > 0$ and $h_0 > 0$ are given.

1. Write down the social planning problem for this economy. Let $v^{\star}(k,h)$ be the value function corresponding to the planning problem.

2. Use first-order conditions to compute the balanced growth path for the economy (in which both types of capital and consumption all grow at a common rate).

3. Make a direct argument to show that:

$$v^{\star}(\lambda k, \lambda h) = \frac{1}{1-\beta}\log(\lambda) + v^{\star}(k,h).$$

4. Formulate the Bellman equation corresponding to the social planning problem in this economy.

5. Solve the value function numerically for the parameter values $\beta = 0.6$, $\alpha = 0.3$, $\delta = 0.6$, $A = 1$, $B = 2$. You can make use of the previous result, which implies that the solution v of the Bellman equation, which coincides with $v^{\star}$, satisfies:

$$v(k,h) = \frac{1}{1-\beta}\log(k) + v\left(1, \frac{h}{k}\right).$$

That is, you only need to compute the one-dimensional value function:

$$\tilde{v}\left(\frac{h}{k}\right) = v\left(1, \frac{h}{k}\right).$$

Limit the state space to the interval $[0.4\bar{h}/\bar{k}, 2\bar{h}/\bar{k}]$, where $\bar{h}/\bar{k}$ is the ratio of human to physical capital on the balanced growth path. Provide graphs of the value and policy functions (i.e., three-dimensional surfaces for varying k and h), and provide time-series graphs comparing outcomes over 10 periods for the same three economies as before: Economy A starts out on the balanced growth path ($h_0/k_0 = \bar{h}/\bar{k}$), Economy B starts out with the same level of physical capital as Economy A, but only half as much human capital, and Economy C starts out with as much human capital as Economy A, but only half as much physical capital. I recommend plotting the time series in logs, so that balanced growth shows up as straight lines.